Multilayer Perceptron

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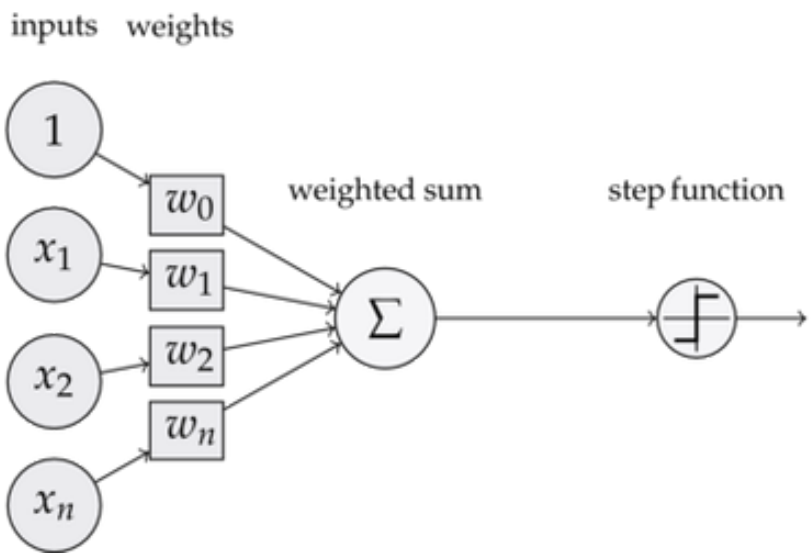
Perceptron

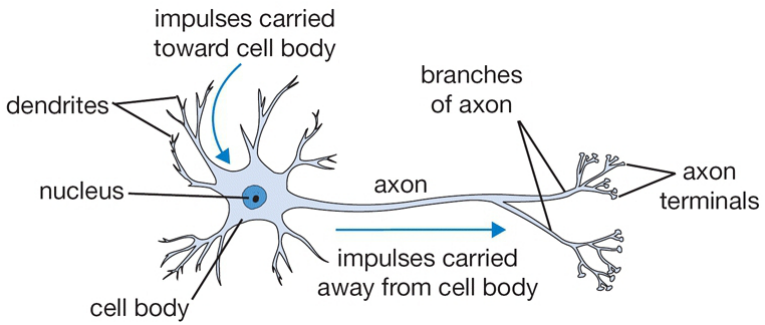
- Perceptron is a single-layer neural network.
- It is a class of linear classifier.



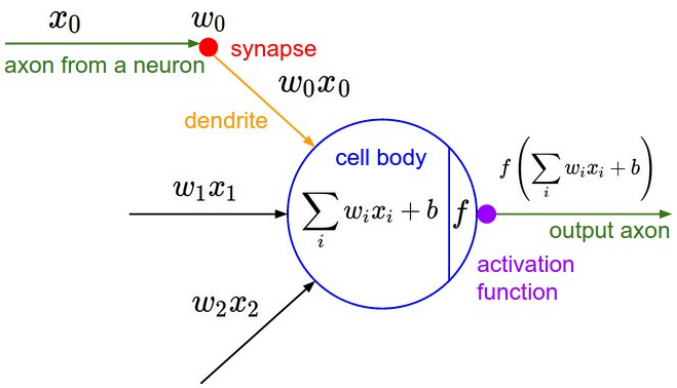
Multilayer Perceptron

- Multilayer perceptron (MLP) is a multi-layer (more than one) neural network.
- It can be a linear classifier with linear activation function and a nonlinear classifier with nonlinear activation function.
- Each node is called to as neuron.
- Neuron is biologically inspired.
- All layers of MLP are fully-connected.

Neuron

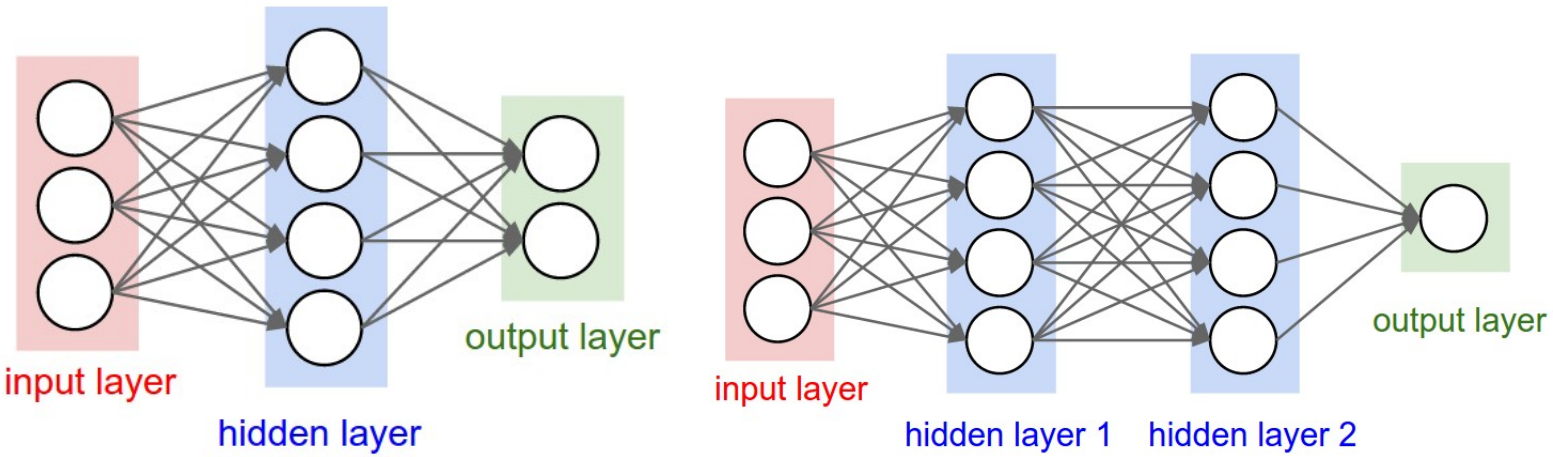

impulses carried
toward cell body
dendrites
branches
of axon
nucleus
axon
axon
terminals
impulses carried
away from cell body
cell body

(a)


x_0
w_0
synapse
axon from a neuron
$w_0 x_0$
dendrite
cell body
$f\left(\sum_i w_i x_i + b\right)$
$w_1 x_1$
$\sum_i w_i x_i + b$
f
output axon
activation
function
$w_2 x_2$

(b)

Multilayer Perceptron



(c) (d)

- (c) can be expressed as

$$f(\mathbf{x}) = \sigma(\mathbf{W}_2^\top \sigma(\mathbf{W}_1^\top \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2) \tag{1}$$

- (d) is

$$f(\mathbf{x}) = \sigma(\mathbf{W}_3^\top \sigma(\mathbf{W}_2^\top \sigma(\mathbf{W}_1^\top \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2) + \mathbf{b}_3) \tag{2}$$

Activation Functions

- Sigmoid.
- Logistic: $\sigma(x) = \frac{1}{1+\exp(-x)}$.
- Hyperbolic tangent: $\tanh(x) = \frac{\exp(x)-\exp(-x)}{\exp(x)+\exp(-x)}$.
- ReLU: $f(x) = \max(0, x)$.
- Leaky ReLU: $f(x) = \mathbb{1}(x < 0)(ax) + \mathbb{1}(x \geq 0)(x)$.

Learning Strategy

- Back-propagation using a chain rule.

Thank you.

References

- http://cs231n.github.io/neural-networks-1/